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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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ECA5610 –ADVANCED WIRELESS COMMUNICATION SYSTEMS / 5G / 6G COMMUNICATION SYSTEMS

LAB REPORT

BY

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Git Hub Link: <https://github.com/ROHITKUMAR013/ECA-5610---5G-6G-COMMUNICATION->



LIST OF EXPERIMENTS – ECA5610

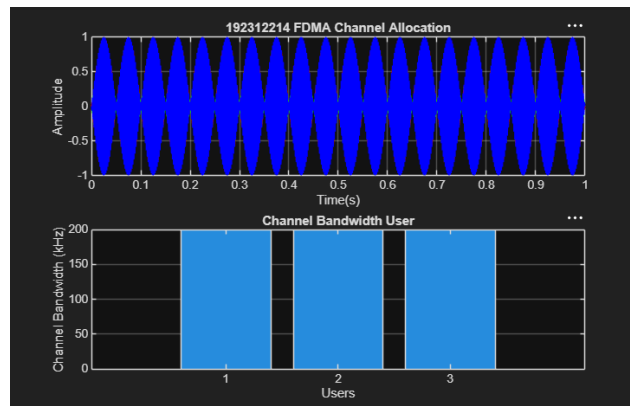
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Ex. No:1: Simulation of FDMA in MATLAB

Objective 1: To simulate an FDMA system by allocating individual frequency channels to multiple users and determine the channel bandwidth assigned to each user.

Program

```
clc; clear; close all;
fs=1000; t=0:1/fs:1;
TBW=600; Users=3;
CBW=TBW/Users;
fc=[100 300 500];
m=sin(2*pi*10*t);
s1=m.*cos(2*pi*fc(1)*t);
s2=m.*cos(2*pi*fc(2)*t);
s3=m.*cos(2*pi*fc(3)*t);
subplot(2,1,1)
plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on
title('192312214 FDMA Channel
Allocation'),xlabel('Time(s)'),ylabel('Amplitude')
subplot(2,1,2)
bar(1:Users,CBW*ones(1,Users)),grid on
xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth
User')
```



Objective 2: To calculate the data rate achieved by each user in the FDMA system based on the allocated channel bandwidth

Program

```
clc; clear; close all;
```

```
BW = 200; % Channel Bandwidth (kHz)
```

```
Users = 3;
```

```
M = 2; % BPSK Modulation
```

```
DataRate = BW*log2(M); % Data Rate (kbps)
```

```
Rate = DataRate*ones(1,Users);
```

```
disp(' User Data Rate (kbps)')
```

```
disp([(1:Users)' Rate'])
```

```
figure
```

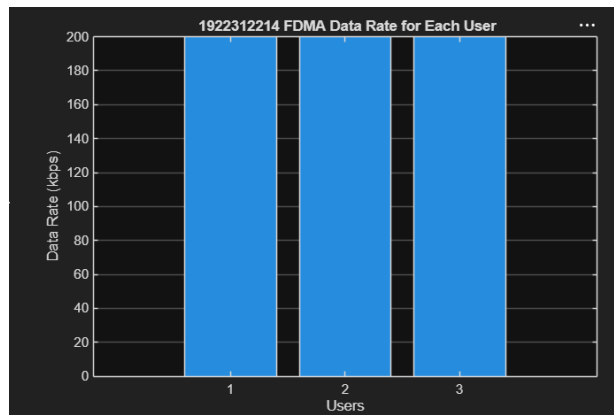
```
bar(1:Users,Rate)
```

```
xlabel('Users')
```

```
ylabel('Data Rate (kbps)')
```

```
title(' 1922312214 FDMA Data Rate for Each User')
```

```
grid on
```



Objective 3: To evaluate the communication quality of the FDMA system by calculating the Signal-to Noise Ratio (SNR) for different users

```
clc; clear; close all;
```

```
Ps = 1; % Signal Power (W)
```

```
Pn = [0.1 0.05 0.01]; % Noise Power (W)
```

```
Users = 1:3;
```

```
SNR = 10*log10(Ps./Pn); % SNR in dB
```

```
disp('User Noise Power(W) SNR(dB)')
```

```
disp([Users' Pn' SNR'])
```

```
figure
```

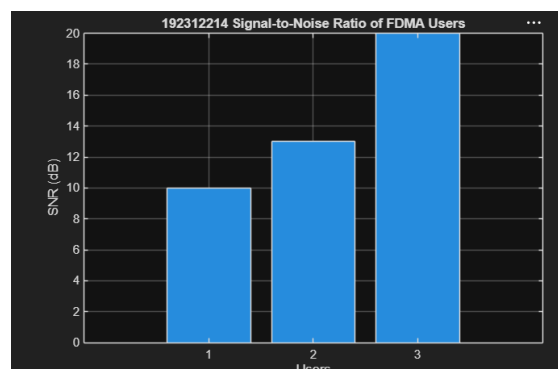
```
bar(Users,SNR)
```

```
xlabel('Users')
```

```
ylabel('SNR (dB)')
```

```
title('192312214 Signal-to-Noise Ratio of FDMA Users')
```

```
grid on
```

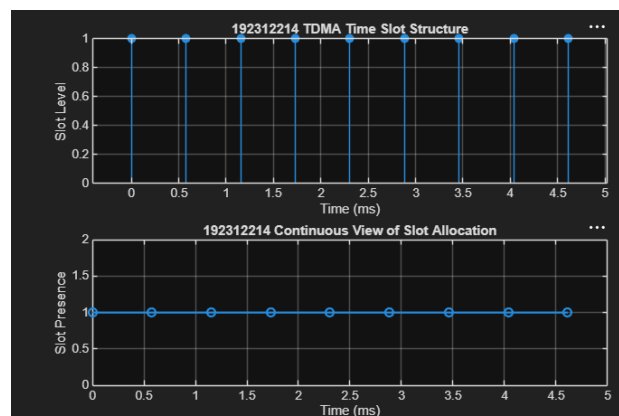


EXP 2: Implementing TDMA Slot Allocation in GSM.

Objective 1: Matlab Code and Output

Program

```
clc; clear; close all;
FrameTime = 4.615; Slots = 8;
SlotDuration = FrameTime/Slots;
t = 0:SlotDuration:FrameTime;
figure
subplot(2,1,1)
stem(t,ones(size(t)),'filled')
title('192312214 TDMA Time Slot Structure')
xlabel('Time (ms)')
ylabel('Slot Level')
grid on
subplot(2,1,2)
plot(t,ones(size(t)),'-o','LineWidth',1.5)
title('192312214 Continuous View of Slot Allocation')
xlabel('Time (ms)')
ylabel('Slot Presence')
grid on
```



Objective 2: Matlab Code and Output

```
Program clc; clear; close all;
```

```
TotalSlots = 8;
```

```
AllocatedSlots = 6;
```

```
FreeSlots = TotalSlots - AllocatedSlots;
```

```
Utilization = (AllocatedSlots/TotalSlots)*100;
```

```
disp('Frame Utilization (%)')
```

```
disp(Utilization)
```

```
figure
```

```
% Graph 1: Simple Line Plot (SAFE - no bar function)
```

```
subplot(2,1,1)
```

```
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
```

```
title('192312214 GSM Frame Utilization (Allocated vs Free)')
```

```
xlabel('Slot Type Index (1=Allocated, 2=Free)')
```

```
ylabel('Number of Slots')
```

```
grid on
```

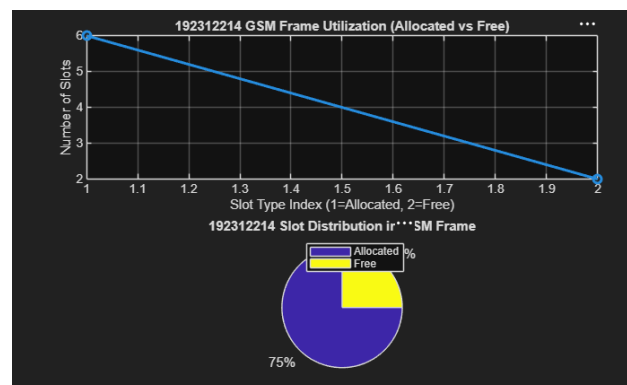
```
% Graph 2: Pie Chart (SAFE)
```

```
subplot(2,1,2)
```

```
pie([AllocatedSlots FreeSlots])
```

```
title('192312214 Slot Distribution in GSM Frame')
```

```
legend('Allocated','Free')
```



Objective 3: Matlab Code and Output

Program

```
clc; clear; close all;

Users = 1:8; % Total users

TotalSlots = 5; % Available slots

Alloc = zeros(1,8); % Initialize allocation

Alloc(1:TotalSlots) = 1; % Assign slots to first users

Blocked = 1 - Alloc; % Remaining users blocked

disp('User Wise Slot Allocation (1=Allocated, 0=Blocked)')

% FIXED: Closed the disp() function properly and simplified the variablename
disp(table(Users', Alloc', 'VariableNames', {'User', 'Status'})) % FIXED: Placed on its own line

% Graph 1: Allocation per user

subplot(2,1,1)

stem(Users, Alloc, 'filled', 'LineWidth', 2)

title('192312214 TDMA Slot Allocation per User')

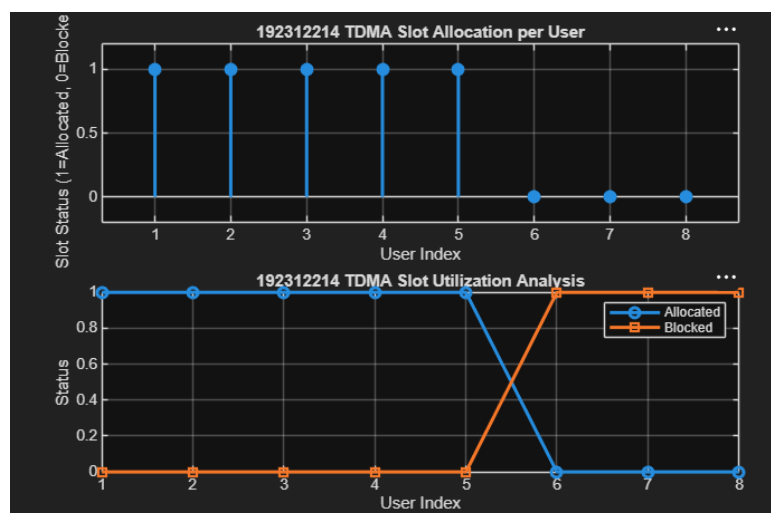
xlabel('User Index')

ylabel('Slot Status (1=Allocated, 0=Blocked)')

ylim([-0.2 1.2])

grid on

output
```

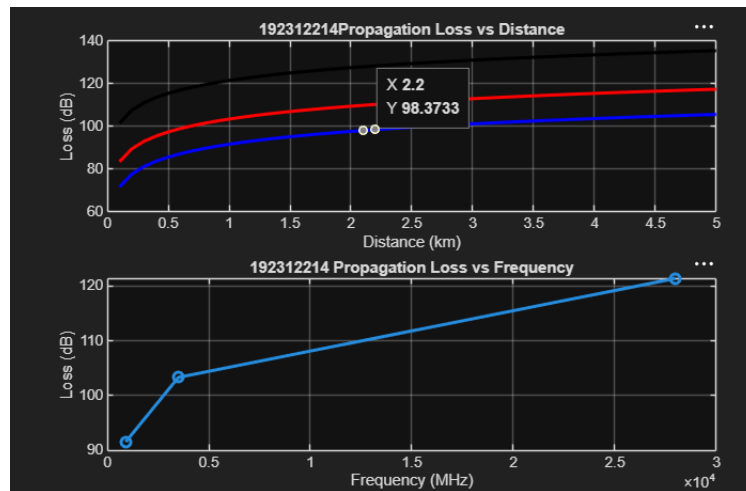


EXP 3: Spectrum Analysis of Sub-1 GHz, Midband & mm Wave by Plotting Propagation Loss.

Objective 1: Matlab Code and Output:

Program

```
clc; clear; close all;
d = 0.1:0.1:5; % Distance (km)
f = [900 3500 28000]; % Frequency (MHz)
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
figure
subplot(2,1,1)
plot(d,PL1,'b','LineWidth',2); hold on
plot(d,PL2,'r','LineWidth',2)
plot(d,PL3,'k','LineWidth',2)
title('192312214 Propagation Loss vs Distance')
xlabel('Distance (km)')
ylabel('Loss (dB)')
grid on
subplot(2,1,2)
Freq = [900 3500 28000];
Loss = [PL1(10) PL2(10) PL3(10)];
plot(Freq,Loss,'-o','LineWidth',2)
title('192312214 Propagation Loss vs Frequency')
xlabel('Frequency (MHz)')
ylabel('Loss (dB)')
grid on
```



Objective 2: Matlab Code and Output:

Program

```
clc; clear; close all;

d = 2; % fixed distance (km)
f = [900 3500 28000]; % MHz
PL = 32.44 + 20*log10(f) + 20*log10(d);

figure

% ----- Graph 1: Frequency vs Loss -----
subplot(2,1,1)
plot(f,PL,'-o','LineWidth',2)
title('192312214 Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on

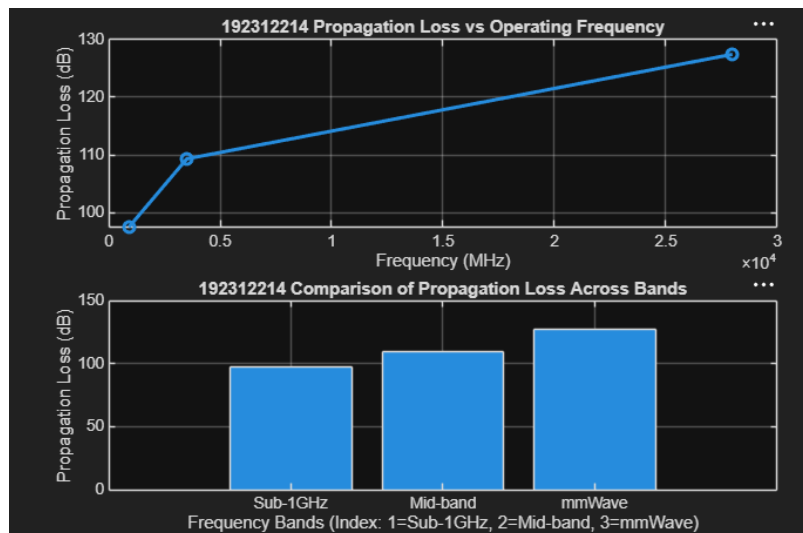
% ----- Graph 2: Band Comparison -----
subplot(2,1,2)
bar(PL)
title('192312214 Comparison of Propagation Loss Across Bands')
xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')
```

```

ylabel('Propagation Loss (dB)')
xticklabels({'Sub-1GHz','Mid-band','mmWave'})
grid on

```

output



Objective 3: Matlab Code and Output:

Program

```

clc; clear; close all;

% Frequency bands (MHz)
f = [900 3500 28000];

% Distance (km)
d = 2;

% Propagation Loss (FSPL)
PL = 32.44 + 20*log10(f) + 20*log10(d);

figure

% ----- Graph 1: Simple Line Plot (SAFE) -----

subplot(2,1,1)

plot(f, PL, '-o', 'LineWidth', 2)

title('192312214 Propagation Loss vs Operating Frequency')

```

```
xlabel('Frequency (MHz)')
```

```
ylabel('Propagation Loss (dB)')
```

```
grid on
```

```
% ----- Graph 2: Safe Comparison Plot -----
```

```
subplot(2,1,2)
```

```
plot(1:3, PL, '-s', 'LineWidth', 2)
```

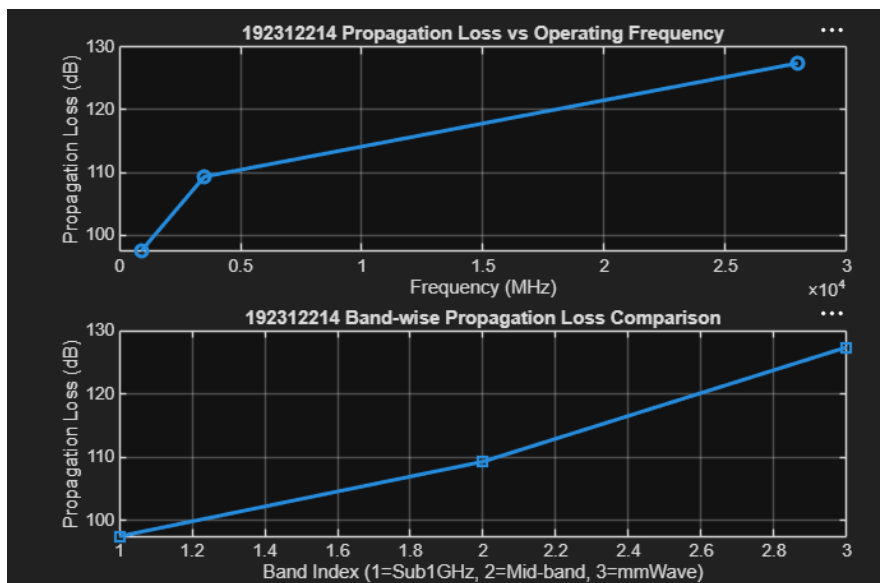
```
title('192312214 Band-wise Propagation Loss Comparison')
```

```
xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')
```

```
ylabel('Propagation Loss (dB)')
```

```
grid on
```

output



EXP 4: OFDMA Resource Block Allocation

Program

```
clc; clear; close all;

% System parameters
bandwidth = 10e6; % 10 MHz
rb_size = 180e3; % 180 kHz per RB
% Total Resource Blocks
total_RB = floor(bandwidth / rb_size);

% Users
users = 1:6;

% RB allocation (simple equal + remainder distribution)
base = floor(total_RB / length(users));
alloc = base * ones(1, length(users));

% FIXED: Moved the +1 to the same line, and moved utilization to a new
linealloc(1:mod(total_RB, length(users))) =
alloc(1:mod(total_RB, length(users))) + 1;

utilization = (alloc / total_RB) * 100;

disp('Total Resource Blocks:')
disp(total_RB)

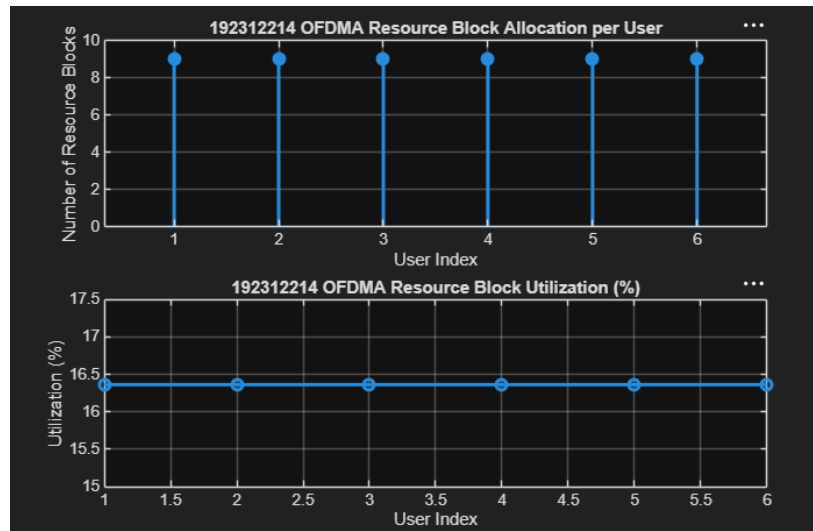
disp('RB Allocation per User:')
disp(alloc)

figure;
subplot(2,1,1)
stem(users, alloc, 'filled', 'LineWidth', 2)
title('192312214 OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on
```

```

subplot(2,1,2)
plot(users, utilization, '-o', 'LineWidth', 2)
title('192312214 OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on

```



Objective 2 Matlab Code and Output:

Program

```

clc; clear; close all;

% Total system resource blocks
total_RB = 50;

% Number of active users
users = 1:5;

% RB allocation to users (example scenario)
alloc = [12 10 9 8 11];

% Resource Block Utilization (%)
util = (alloc / total_RB) * 100;

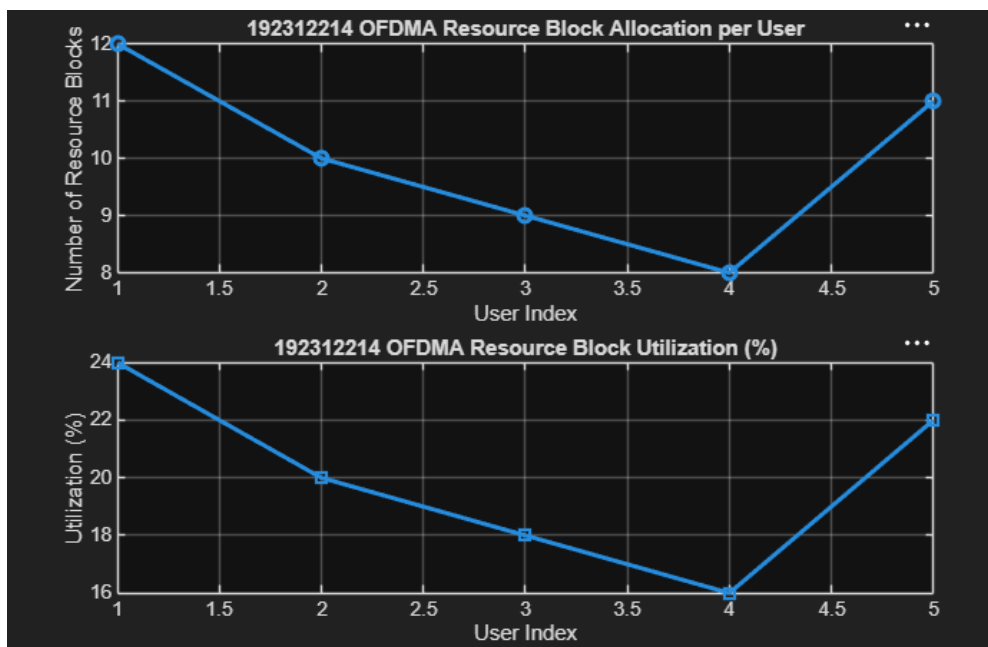
disp('Resource Block Utilization (%) per User:')

```

```

disp(util')
figure
subplot(2,1,1)
plot(users, alloc, '-o', 'LineWidth', 2)
title('192312214 OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on
subplot(2,1,2)
plot(users, util, '-s', 'LineWidth', 2)
title('192312214 OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on

```



Objective 3 Matlab Code and Output

Program

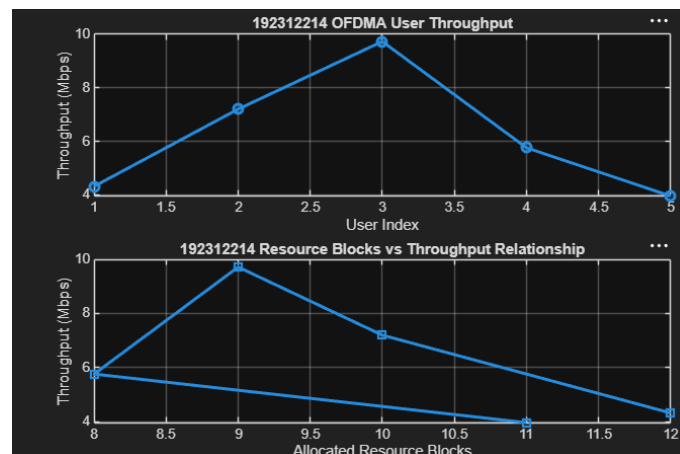
```
clc; clear; close all;
```

```

% System parameters
rb_bw = 180e3; % Bandwidth per RB (180 kHz)
users = 1:5;
alloc_RB = [12 10 9 8 11];
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
disp('User Throughput (Mbps):')
disp(throughput)

figure
subplot(2,1,1)
title('192312214 OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on
subplot(2,1,2)
plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
title('192312214 Resource Blocks vs Throughput Relationship')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on

```



EXP 5: MIMO System Simulation with BER Analysis

Objective 1 Matlab Code and Output

Program

```
clc; clear; close all;

% SNR range
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);

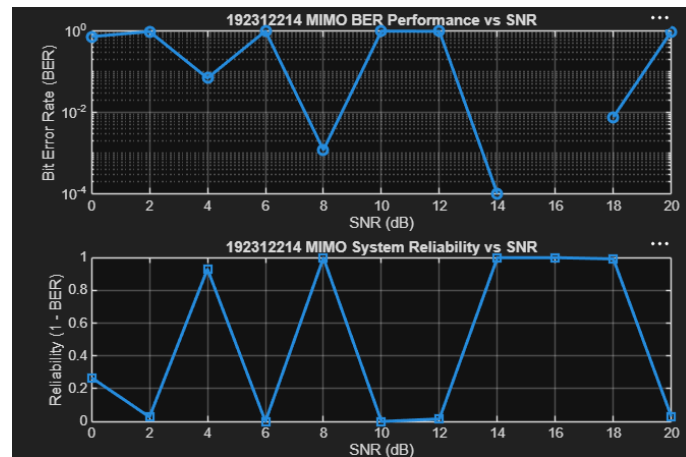
% MIMO parameters
Nt = 2; Nr = 2;
BER = zeros(1,length(SNR));

for i = 1:length(SNR)
    bits = randi([0 1],10000,1);
    symbols = 2*bits-1; % BPSK
    noise = (randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i)); tx = H(1,1)*symbols' + noise(1,:);
    rx = real(tx) > 0;
    BER(i) = sum(rx' ~= bits)/length(bits);
end

figure
subplot(2,1,1)
semilogy(SNR_dB,BER,'-o','LineWidth',2)
title('192312214 MIMO BER Performance vs SNR')
xlabel('SNR (dB)')
ylabel('Bit Error Rate (BER)')

subplot(2,1,2)
plot(SNR_dB,1-BER,'-s','LineWidth',2)
title('192312214 MIMO System Reliability vs SNR')
xlabel('SNR (dB)')
ylabel('Reliability (1 - BER)')

grid on
```



Objective 2 Matlab Code and Output

Program

```
clc; clear; close all;

% SNR range (dB)
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);

% MIMO parameters
Nt = 2; Nr = 2;

% Simulated reliability metric (inverse BER model)
Reliability = zeros(1,length(SNR));

for i = 1:length(SNR)
    % simple channel model effect
    H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
    noise_power = 1/SNR(i);
    % reliability increases with SNR
    Reliability(i) = 1 - exp(-SNR(i)/10);
end

figure
subplot(2,1,1)
plot(SNR_dB, Reliability, '-o', 'LineWidth', 2)
title('MIMO Reliability vs SNR')
xlabel('SNR (dB)')
```

```
ylabel('Reliability (0 to 1)')
```

```
grid on
```

```
subplot(2,1,2)
```

```
plot(SNR_dB, noise_power*ones(size(SNR_dB)), '-s', 'LineWidth', 2)
```

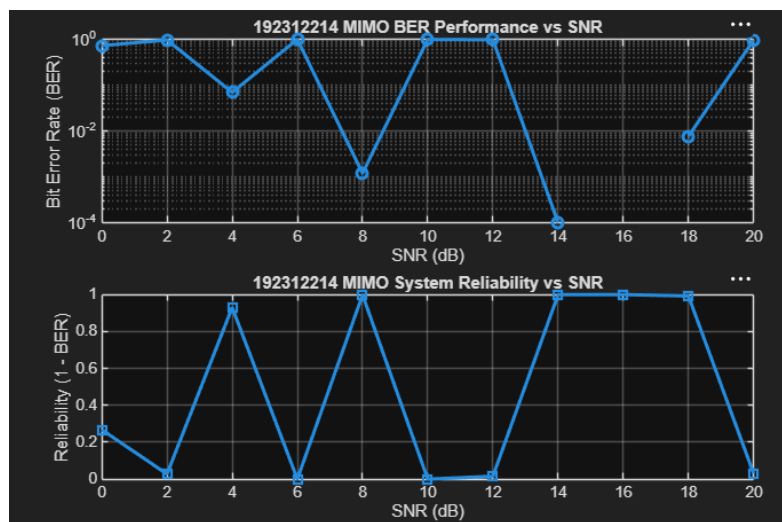
```
title('Noise Power Variation with SNR')
```

```
xlabel('SNR (dB)')
```

```
ylabel('Noise Power')
```

```
grid on
```

```
output
```



Objective 3 Matlab Code and Output:

Program

```
clc; clear; close all;
```

```
% Resource Block parameters
```

```
rb_bw = 180e3; % 180 kHz per RB
```

```
% Users
```

```
users = 1:5;
```

```
% Allocated Resource Blocks per user
```

```
alloc_RB = [10 12 8 15 5];
```

```
% Modulation efficiency (bits/symbol)
```

```
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
```

```
% Throughput calculation (Mbps)
```

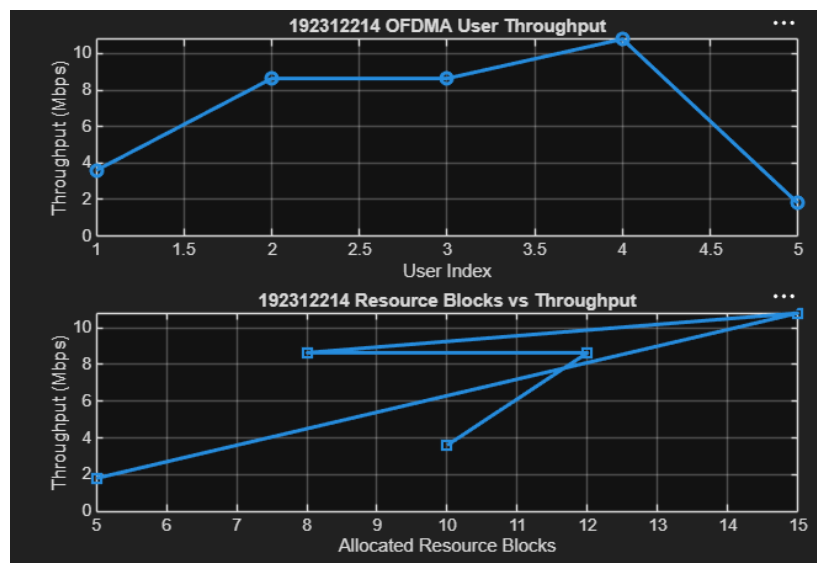
```
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
```

```

% Display results
disp('User Throughput (Mbps):')
disp(throughput')
figure
% ----- Graph 1: Throughput per User -----
subplot(2,1,1)
plot(users, throughput, '-o','LineWidth',2)
title('192312214 OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on
% ----- Graph 2: RB vs Throughput -----
subplot(2,1,2)
plot(alloc_RB, throughput, '-s','LineWidth',2)
title('192312214 Resource Blocks vs Throughput')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on

```

output



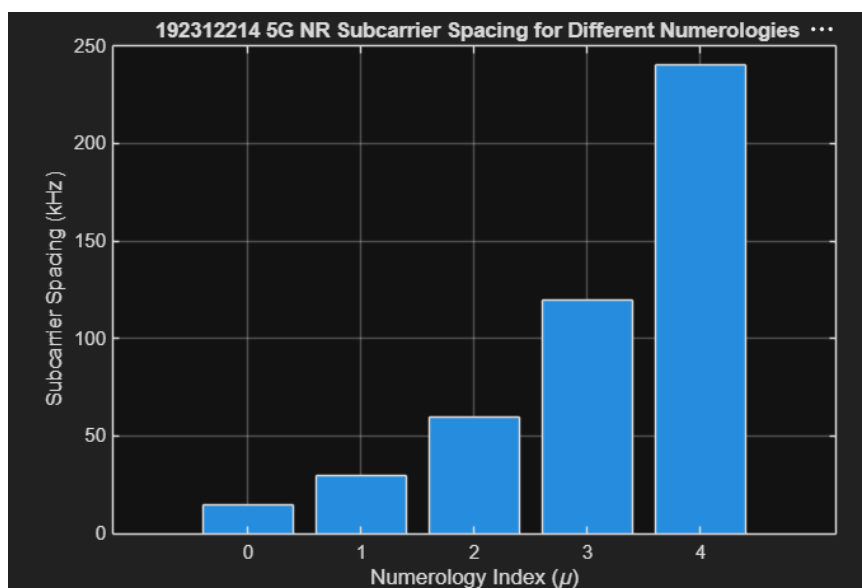
EXP: 6 5G NR Frame Structure Analysis with Different Subcarrier Spacing

Objective – 1: Analyze Different Subcarrier Spacing Values

Program code:

```
clc;
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
mu = 0:4; % Numerology Index
bar(scs)
grid on
set(gca,'XTick',1:5)
set(gca,'XTickLabel',mu)
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('192312214 5G NR Subcarrier Spacing for Different Numerologies')
```

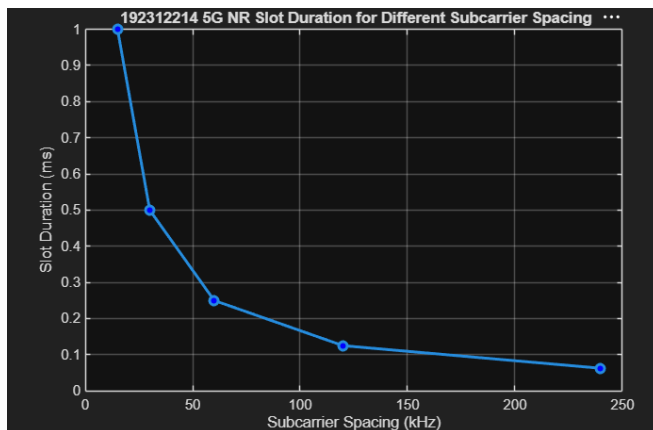
output:



Objective – 2 Compare Slot Duration

Program code:

```
clear;
close all;
scs = [15 30 60 120 240];
slot = [1 0.5 0.25 0.125 0.0625];
figure;
plot(scs, slot, 'o-', 'LineWidth', 2, 'MarkerFaceColor', 'b')
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Slot Duration (ms)')
title('192312214 5G NR Slot Duration for Different Subcarrier Spacing')
```



Objective – 3 Study the Number of OFDM Symbols

Program code:

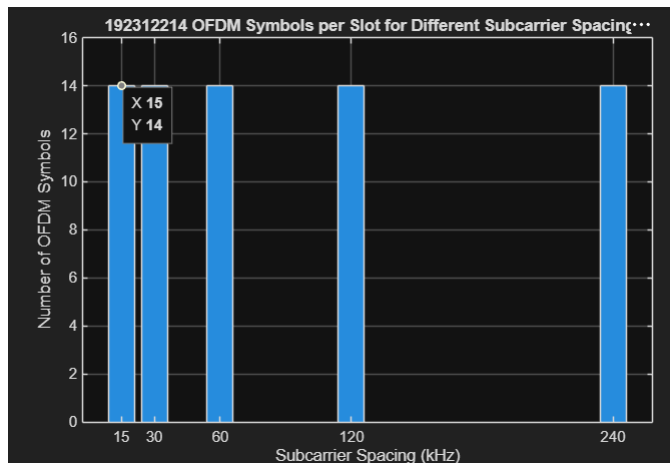
```
clc;
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
symbols = [14 14 14 14 14]; % OFDM Symbols per Slot
```

```

figure
bar(scs, symbols)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Number of OFDM Symbols')
title('192312214 OFDM Symbols per Slot for Different Subcarrier Spacing')
ylim([0 16])

```

output:



EXP :7 Analog Beamforming Analysis Using MATLAB

Objective – 1: Analyze Beamforming Gain (dB)

Program code:

```

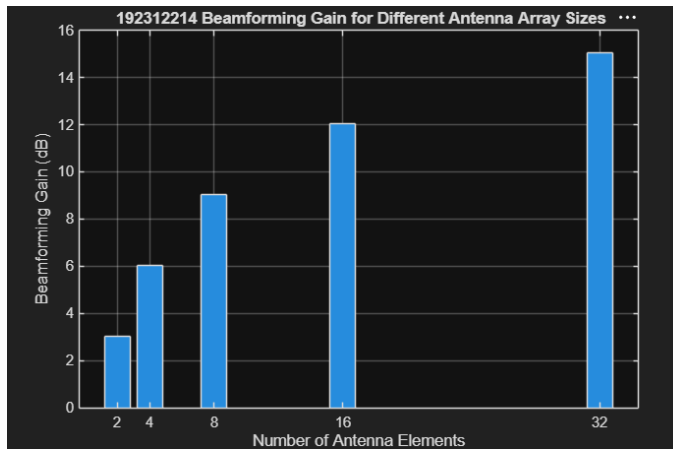
clc;
clear;
close all;
antennas = [2 4 8 16 32]; % Number of Antenna Elements
gain = 10*log10(antennas); % Beamforming Gain (dB)
figure
bar(antennas, gain)
grid on
xlabel('Number of Antenna Elements')

```

```
ylabel('Beamforming Gain (dB)')
```

```
title('192312214 Beamforming Gain for Different Antenna Array Sizes')
```

output:



Objective – 2: Compare Beam Width (Degrees)

Program code:

```
clc;
```

```
clear;
```

```
close all;
```

```
antennas = [2 4 8 16 32]; % Number of Antenna Elements
```

```
beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width (degrees)
```

```
figure
```

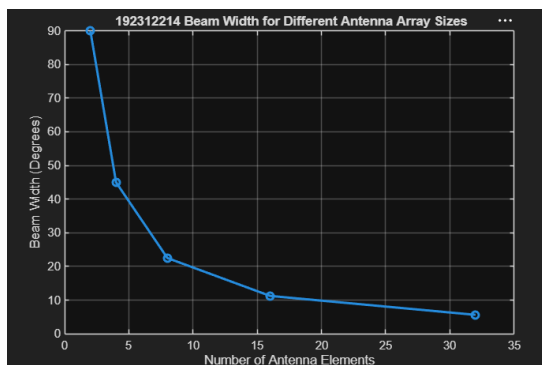
```
plot(antennas, beamwidth,'o-','LineWidth',2)
```

```
grid on
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Beam Width (Degrees)')
```

```
title('192312214 Beam Width for Different Antenna Array Sizes')
```

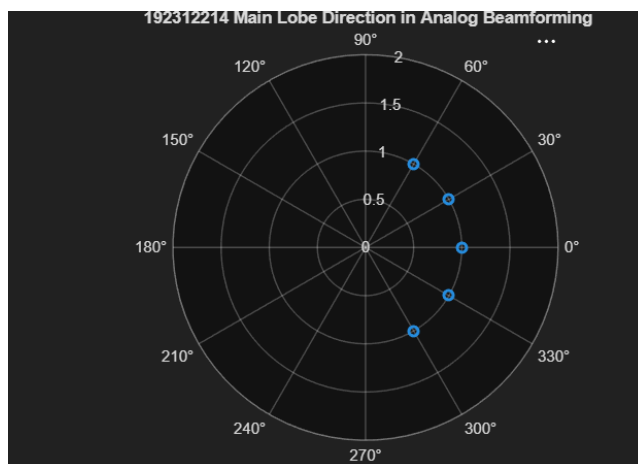


Objective – 3: Analyze Main Lobe Direction (Degrees)

program code:

```
clc;
clear;
close all;
theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)
gain = [1 1 1 1 1]; % Constant normalized gain
figure
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
title(' 192312214 Main Lobe Direction in Analog Beamforming')
```

output:

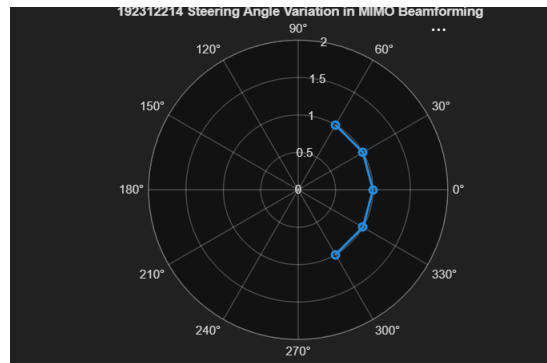


EXP 8: Steering Angle Variation in MIMO Beamforming Using MATLAB

Objective – 1 Analyze Steering Angle (Degrees)

Program code:

```
clc;
clear;
close all;
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)
gain = ones(size(angle)); % Normalized Gain
figure
polarplot(deg2rad(angle), gain, 'o-', 'LineWidth', 2)
title(' 192312214 Steering Angle Variation in MIMO Beamforming')
```



Objective – 2 Compare Array Gain (dB)

Program code:

```
clc;
clear;
close all;

angle = [-60 -30 0 30 60]; % Steering Angles (degrees)

gain = [9 12 15 12 9]; % Array Gain (dB)

figure
bar(angle, gain)

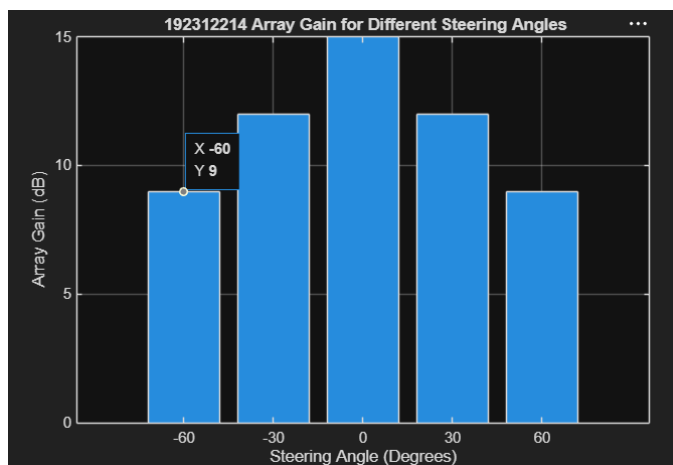
grid on

xlabel('Steering Angle (Degrees)')

ylabel('Array Gain (dB)')

title('192312214 Array Gain for Different Steering Angles')
```

output :



Objective – 3 Evaluate Beam Direction Error (Degrees)

Program code:

```
clc;

clear;

close all;

desired = [-60 -30 0 30 60];
actual = [-58 -32 1 29 62];

error = abs(desired - actual);

figure

plot(desired, error, 'o-', 'LineWidth', 2)

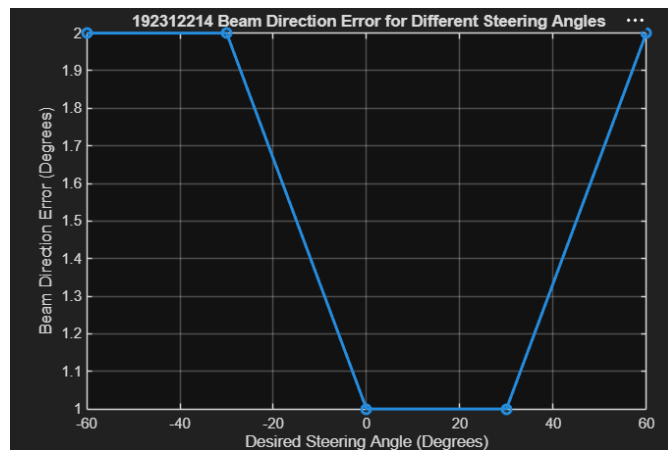
grid on

xlabel('Desired Steering Angle (Degrees)')

ylabel('Beam Direction Error (Degrees)')

title('192312214 Beam Direction Error for Different Steering Angles')

output :
```



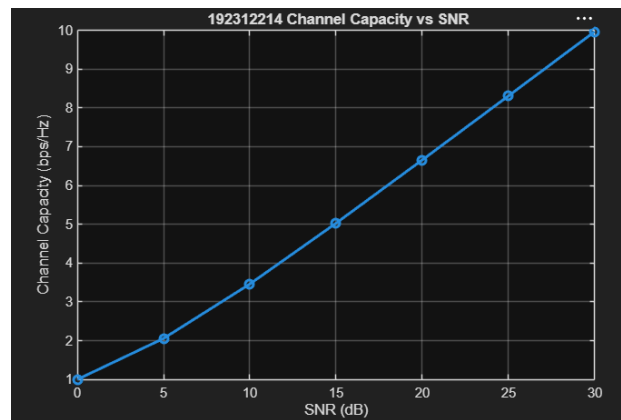
EXP 9: Massive MIMO Capacity vs SNR and Gain Using MATLAB

Objective – 1 Plot of Channel Capacity (bps/Hz) versus SNR(dB).

Program code:

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('192312214 Channel Capacity vs SNR');
```

output



Objective – 2 Bar graph comparing SNR values under different communication scenarios.

Program code:

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
```

```

plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

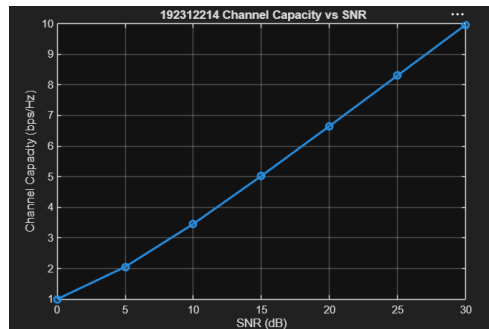
grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title(' 192312214 Channel Capacity vs SNR');

```



Objective – 3 Bar graph showing Antenna Gain for different antenna array sizes.

Program code:

```

clc;

clear;

close all;

antennas = [4 8 16 32 64];

gain = 10*log10(antennas);

figure

bar(antennas, gain)

grid on

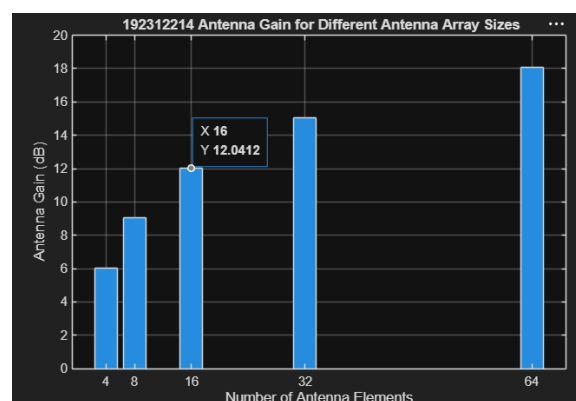
xlabel('Number of Antenna Elements')

ylabel('Antenna Gain (dB)')

title(' 192312214 Antenna Gain for Different Antenna Array Sizes')

```

output



EXP 10: 5G Network Slicing Environment Using MATLAB

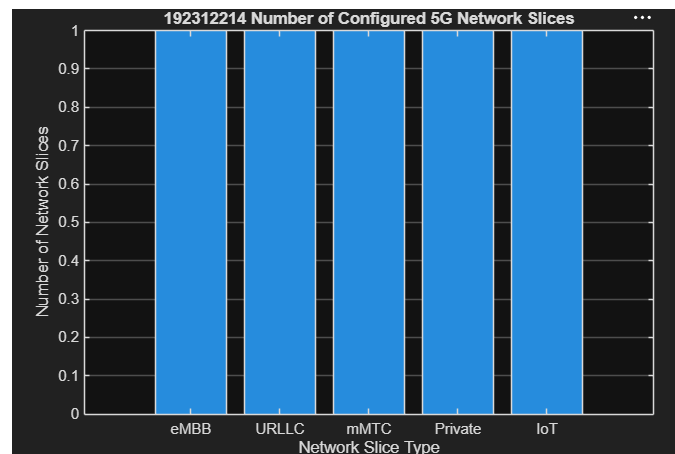
Objective – 1: Analyze the Number of Network Slices

Program code:

```
clc;
clear;
close all;

slices = {'eMBB','URLLC','mMTC','Private','IoT'};
count = [1 1 1 1 1]; % One instance of each network slice

figure
bar(count)
grid on
set(gca,'XTickLabel',slices)
xlabel('Network Slice Type')
ylabel('Number of Network Slices')
title('192312214 Number of Configured 5G Network Slices')
```



Objective – 2: Compare Slice Utilization (%)

Program code:

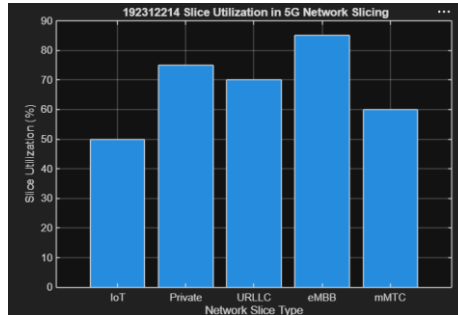
```
clc;
clear;
close all;

slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});utilization = [85 70 60 75 50];

figure;
bar(slices, utilization);
grid on;
```

```
xlabel('Network Slice Type');
ylabel('Slice Utilization (%)');
title(' 192312214 Slice Utilization in 5G Network Slicing');
```

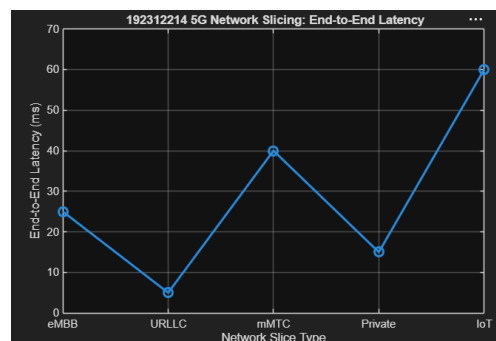
output



Objective – 3: Analyze End-to-End Latency (ms)

Program code:

```
clc;
clear;
close all;
slices = [1 2 3 4 5];
latency = [25 5 40 15 60];
figure;
plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);
grid on;
xticks(slices);
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});
xlabel('Network Slice Type');
ylabel('End-to-End Latency (ms)');
title(' 192312214 5G Network Slicing: End-to-End Latency');
ylim([0 70]);
```



EXP 11: Implementation of IoT-based Smart Agriculture using MATLAB

Objective 1 Matlab Code and Output

Program code:

```
clc; clear; close all;

% Simulated soil moisture data (%)

time = 1:10;

soil_moisture = [30 35 28 25 20 18 22 30 40 45]; threshold = 40;

figure subplot(2,1,1)

plot(time, soil_moisture, '-o', 'LineWidth', 2)

title('192312214 Soil Moisture Monitoring in IoT Smart Agriculture')

xlabel('Time (Hours)')

ylabel('Soil Moisture (%)')

grid on

hold on

yline(threshold, 'r--', 'Threshold')

status = soil_moisture >= threshold;

subplot(2,1,2)

stem(time, status, 'filled', 'LineWidth', 2)

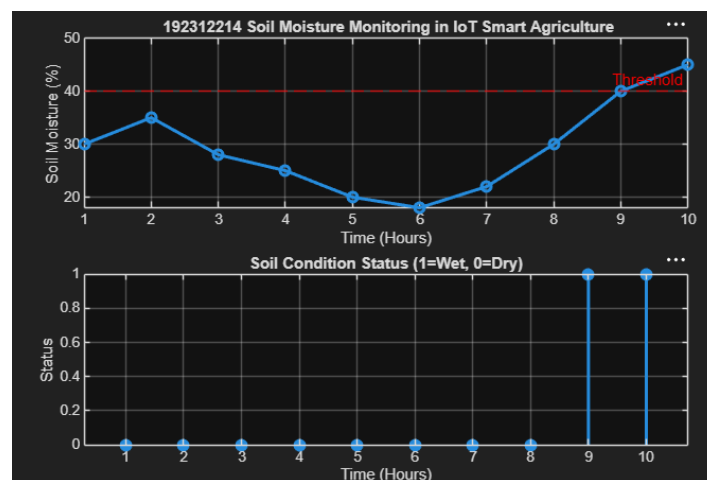
title('Soil Condition Status (1=Wet, 0=Dry)')

xlabel('Time (Hours)')

ylabel('Status')

grid on
```

output



Objective 2 Matlab Code and Output

Program code:

```
clc; clear; close all;

% Simulated soil moisture (%)

time = 1:10;

soil_moisture = [30 35 28 25 20 18 22 30 40 45];

threshold = 40;

pump_status = soil_moisture < threshold; subplot(2,1,1)

plot(time, soil_moisture, '-o','LineWidth',2)

title('192312214 Soil Moisture vs Pump Control')

xlabel('Time (Hours)')

ylabel('Soil Moisture (%)')

grid on hold on

yline(threshold,'r--','Threshold')

subplot(2,1,2)

stem(time, pump_status, 'filled','LineWidth',2)

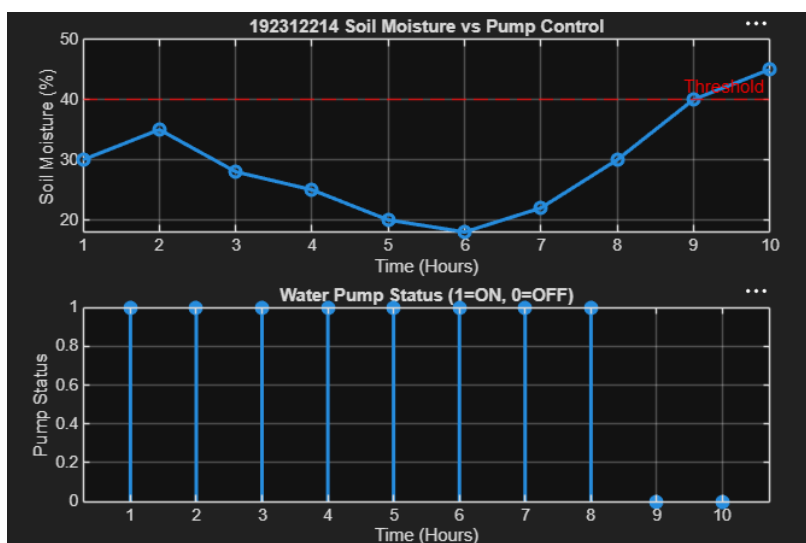
title('Water Pump Status (1=ON, 0=OFF)')

xlabel('Time (Hours)')

ylabel('Pump Status')

grid on
```

Output



Objective 3 :Matlab Code and Output

Program code:

```
clc; clear; close all;

time = 1:10;

soil_moisture = [30 35 28 25 20 18 22 30 40 45];

threshold = 40;

efficiency = (soil_moisture ./ threshold) * 100;

efficiency(efficiency > 100) = 100; % limit to 100%figure

subplot(2,1,1)

plot(time, soil_moisture, '-o', 'LineWidth', 2)

title(' 192312214 Soil Moisture Variation')

xlabel('Time (Hours)')

ylabel('Soil Moisture (%)')

grid onhold on

yline(threshold, 'r--', 'Threshold')

subplot(2,1,2)

plot(time, efficiency, '-s', 'LineWidth', 2)

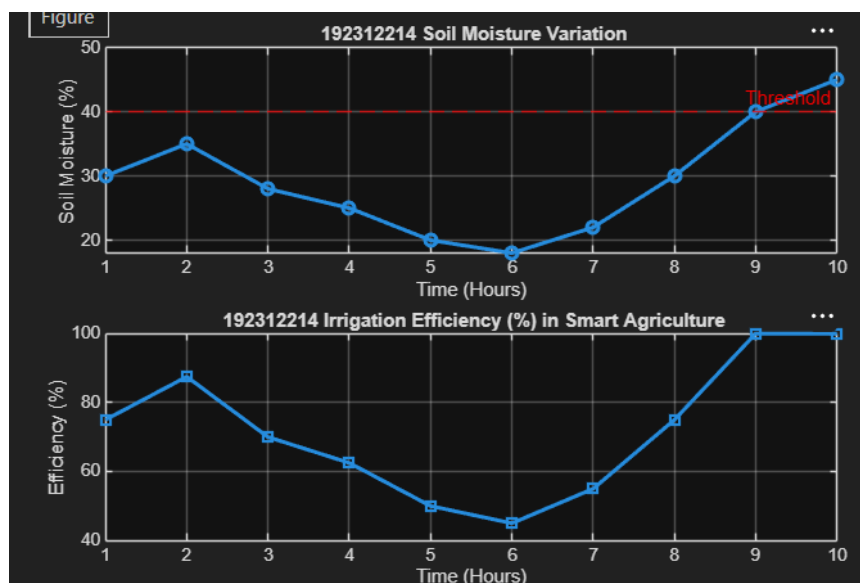
title(' 192312214 Irrigation Efficiency (%) in Smart Agriculture')

xlabel('Time (Hours)')

ylabel('Efficiency (%)')

grid on
```

Output



EXP 12: NB-IoT vs LTE-M Throughput Simulation Date

Objective 1 Matlab Code and Output

Program code:

```
clc; clear; close all;

SNR = 0:5:30;

% Simulated Throughput (Mbps)
NB_IoT = 0.05*(1-exp(-SNR/8));
LTE_M = 1.2*(1-exp(-SNR/8));

disp('SNR (dB) NB-IoT(Mbps) LTE-M(Mbps)')

disp([SNR' NB_IoT' LTE_M'])

subplot(2,1,1)

plot(SNR,NB_IoT,'-o','LineWidth',2)

title('192312214 NB-IoT Throughput vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Throughput (Mbps)')

subplot(2,1,2)

plot(SNR,LTE_M,'-s','LineWidth',2)

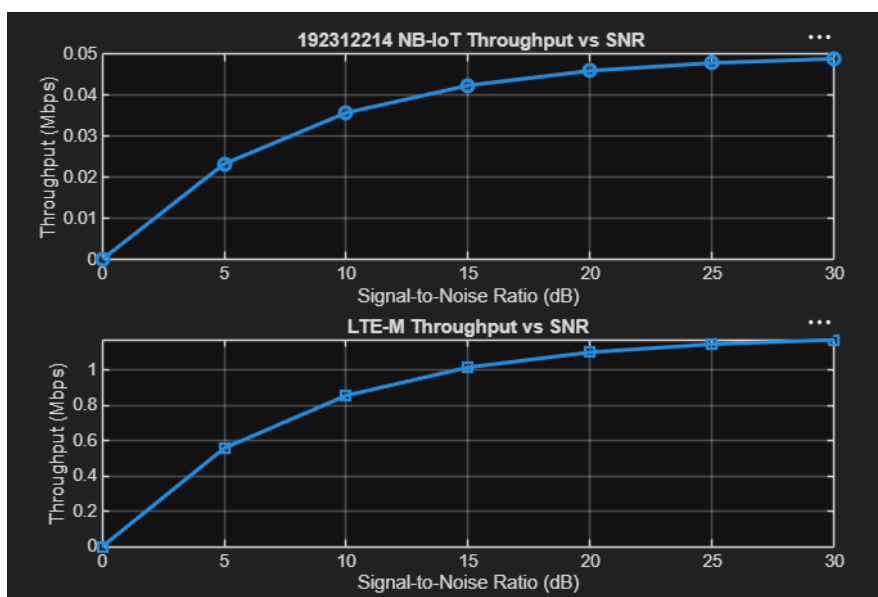
title('LTE-M Throughput vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Throughput (Mbps)')

grid on
```

Output



Objective 2 Matlab Code and Output

```
clc; clear; close all;

SNR = 0:5:30;

% Simulated Latency (ms)
NB_IoT = [180 160 145 130 120 110 100];
LTE_M = [90 80 70 60 50 45 40];

disp('SNR(dB) NB-IoT Latency(ms) LTE-M Latency(ms)')

disp([SNR' NB_IoT' LTE_M'])

subplot(2,1,1)

plot(SNR,NB_IoT,'-o','LineWidth',2)

title('192312214 NB-IoT Latency vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Latency (ms)')grid on

subplot(2,1,2)

plot(SNR,LTE_M,'-s','LineWidth',2)

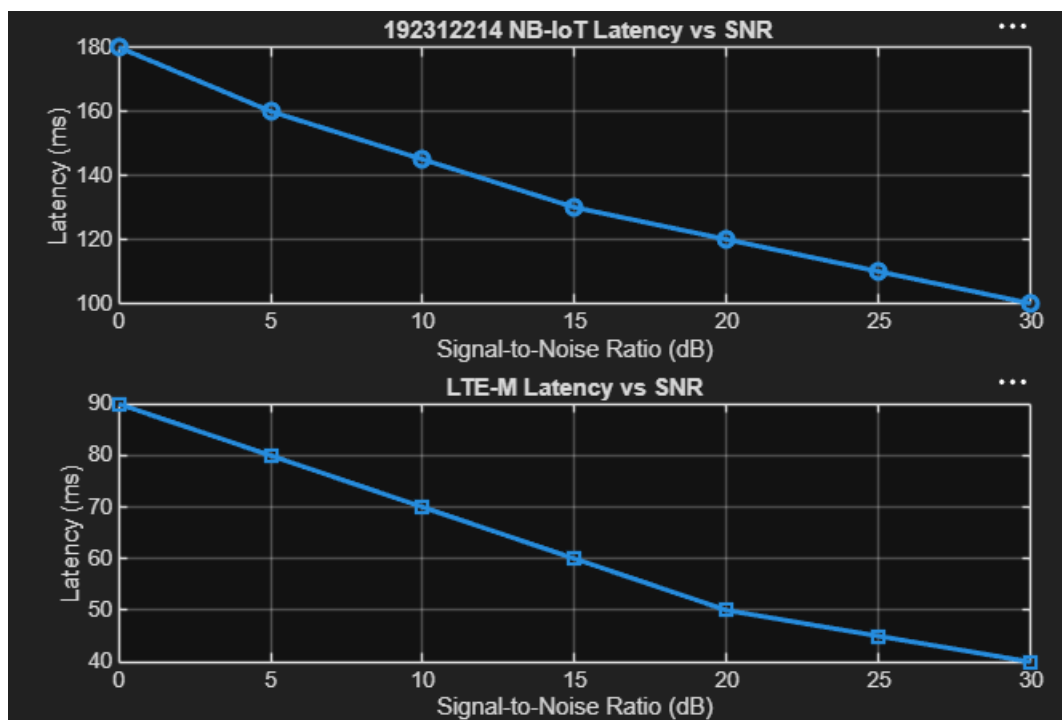
title('LTE-M Latency vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Latency (ms)')

grid on
```

Output:



Objective 3 Matlab Code and Output:

Program code

```
clc; clear; close all;

SNR = 0:5:30;

NB_IoT = [82 86 89 92 95 97 99];

LTE_M = [88 91 94 96 98 99 100];

disp('SNR(dB) NB-IoT PDR(%) LTE-M PDR(%))')

disp([SNR' NB_IoT' LTE_M'])

subplot(2,1,1)

plot(SNR,NB_IoT,'-o','LineWidth',2)

title('192312214 NB-IoT Packet Delivery Ratio vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Packet Delivery Ratio (%)')

grid on

subplot(2,1,2)

plot(SNR,LTE_M,'-s','LineWidth',2)

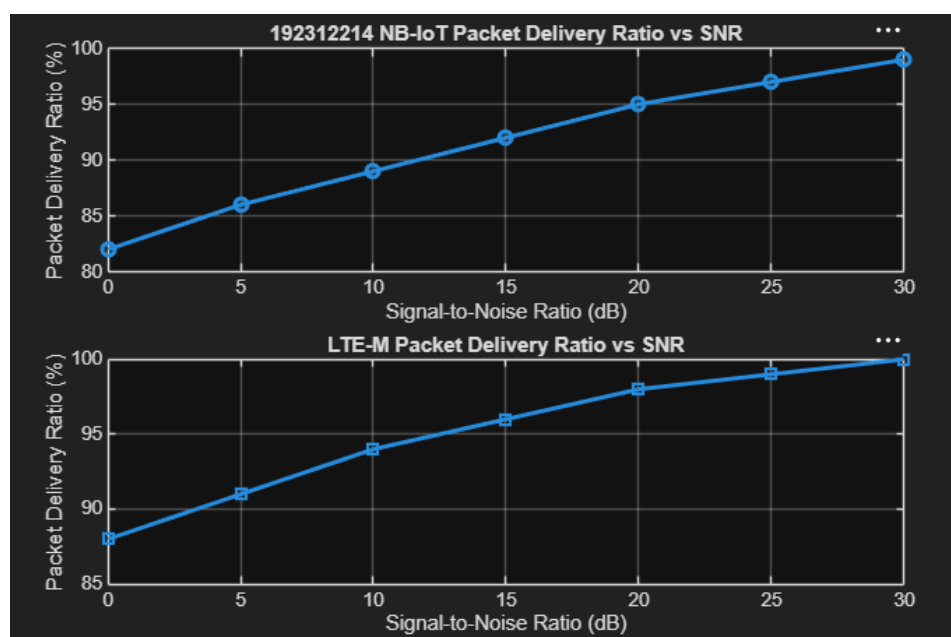
title('LTE-M Packet Delivery Ratio vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Packet Delivery Ratio (%)')

grid on
```

Output:



EXP 13: V2X Latency Comparison Date

Objective 1 Matlab Code and Output

```
clc;clear;close all;

% Vehicle Speed (km/h)

speed = 20:20:120;

% Simulated End-to-End Latency (ms)

latency = [18 16 14 12 10 9];

disp('Vehicle Speed (km/h) End-to-End Latency (ms)')

disp([speed' latency'])

subplot(2,1,1)

plot(speed,latency,'-o','LineWidth',2)

title('192312214 Vehicle Speed vs End-to-End Latency')

xlabel('Vehicle Speed (km/h)')

ylabel('Latency (ms)')

subplot(2,1,2)

plot(1:length(latency),latency,'-s','LineWidth',2)

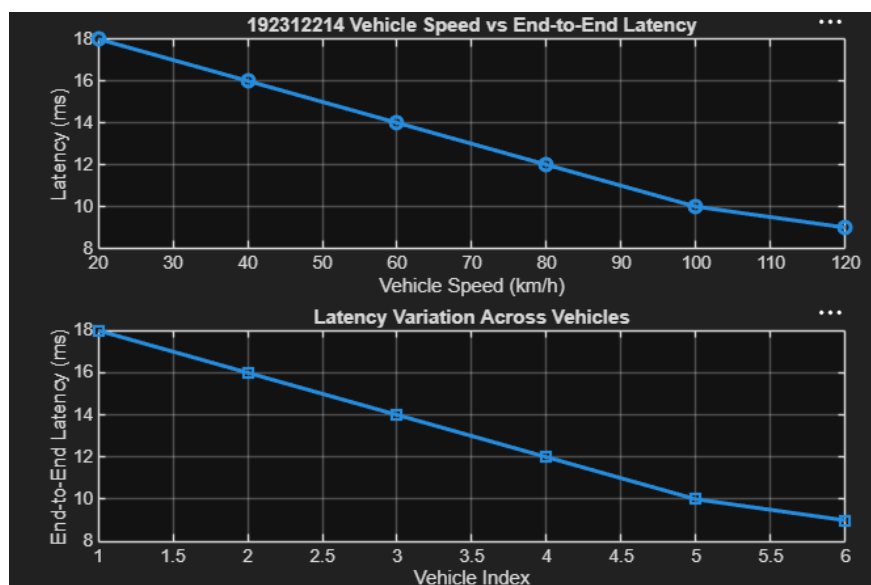
title('Latency Variation Across Vehicles')

xlabel('Vehicle Index')

ylabel('End-to-End Latency (ms)')

grid on
```

Output:



Objective 2 Matlab Code and Output:

Program code:

```
clc; clear; close all;

speed = 20:20:120;

% Simulated Packet Delivery Ratio (%)

PDR = [99 98 97 95 93 90];

disp('Vehicle Speed (km/h) Packet Delivery Ratio (%)')

disp([speed' PDR'])

subplot(2,1,1)

plot(speed,PDR,'-o','LineWidth',2)

title('192312214 Vehicle Speed vs Packet Delivery Ratio')

xlabel('Vehicle Speed (km/h)')

ylabel('Packet Delivery Ratio (%)')

subplot(2,1,2)

plot(1:length(PDR),PDR,'-s','LineWidth',2)

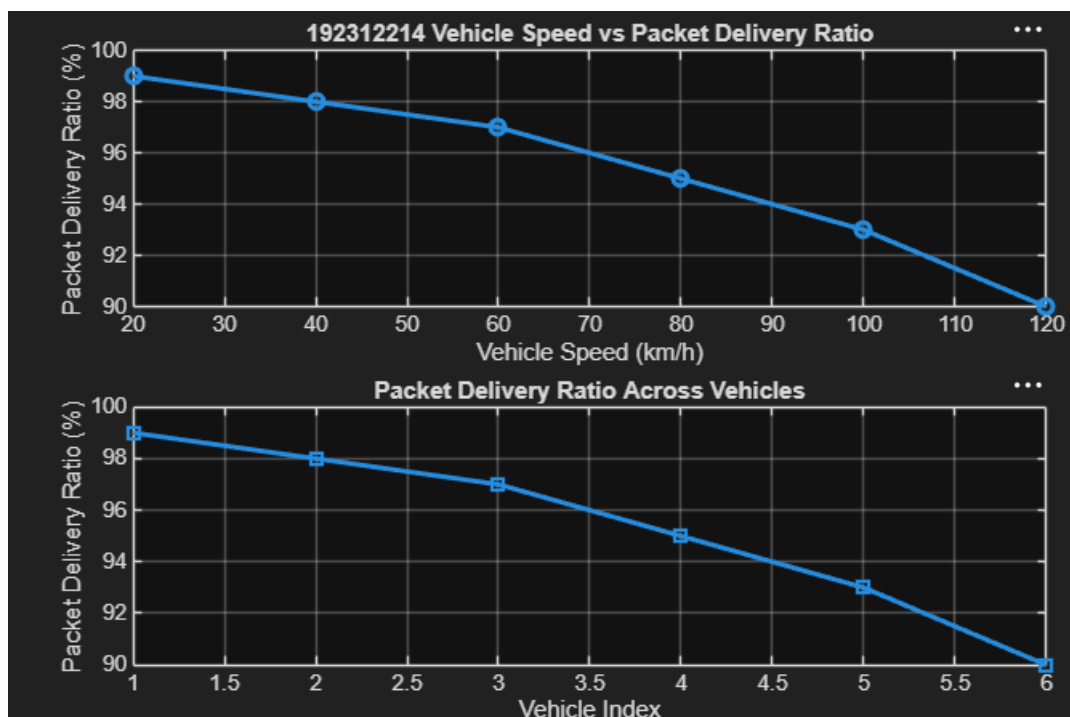
title('Packet Delivery Ratio Across Vehicles')

xlabel('Vehicle Index')

ylabel('Packet Delivery Ratio (%)')

grid on

Output :
```



EXP 14: 5G Throughput vs SNR with Modulation Labels Date

Objective 1 Matlab Code and Output

Program code:

```
clc; clear; close all;

% Vehicle Speed (km/h)

speed = 20:20:120;

PDR = [99 98 97 95 93 90];

disp('Vehicle Speed (km/h) Packet Delivery Ratio (%)')

disp([speed' PDR'])

subplot(2,1,1)

plot(speed,PDR,'-o','LineWidth',2)

title('192312214 Vehicle Speed vs Packet Delivery Ratio')

xlabel('Vehicle Speed (km/h)')

ylabel('Packet Delivery Ratio (%)')subplot(2,1,2)

plot(1:length(PDR),PDR,'-s','LineWidth',2)

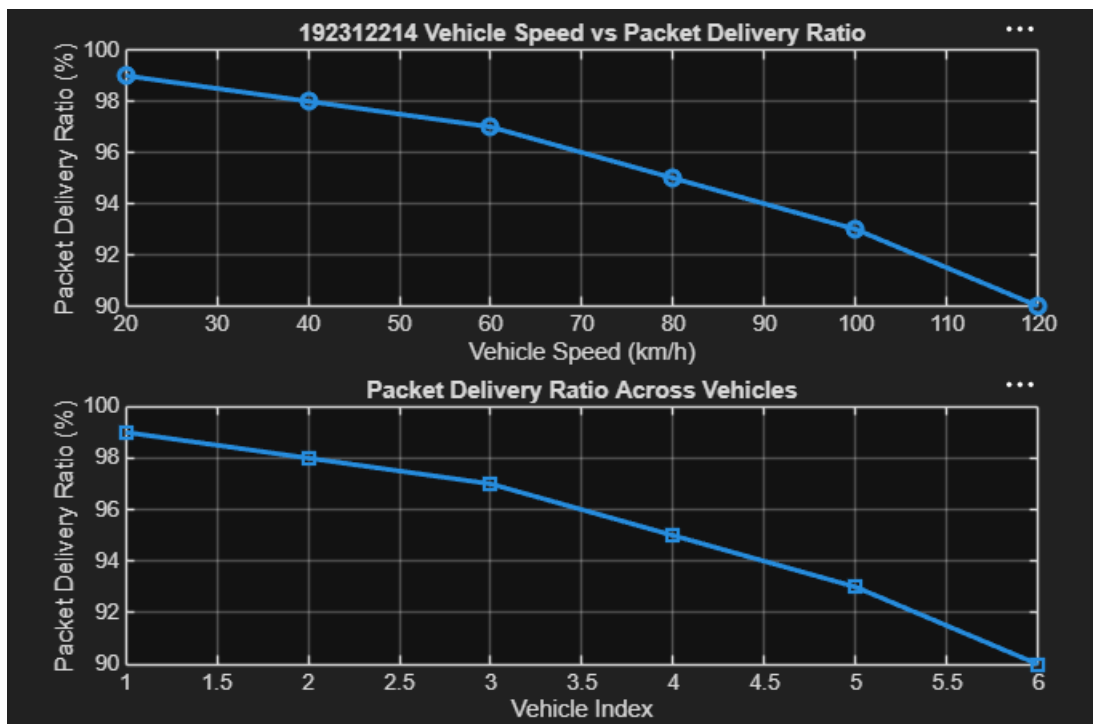
title('Packet Delivery Ratio Across Vehicles')

xlabel('Vehicle Index')

ylabel('Packet Delivery Ratio (%)')

grid on
```

Output



Objective 3 Matlab Code and Output

Program code:

```
clc; clear; close all;

Modulation = {'QPSK','16-QAM','64-QAM','256-QAM'};

SNR = [5 10 18 25];

Efficiency = [2 4 6 8];

disp('Required SNR (dB) and Spectral Efficiency')

disp(table(Modulation',SNR',Efficiency', ...

    'VariableNames',{'Modulation','SNR_dB','Efficiency'})))

subplot(2,1,2)

plot(SNR,Efficiency,'-o','LineWidth',2)

title('192312214 SNR vs Spectral Efficiency')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Spectral Efficiency (bits/s/Hz)')

subplot(2,1,1)

plot(1:4,SNR,'-s','LineWidth',2)xticklabels(Modulation)

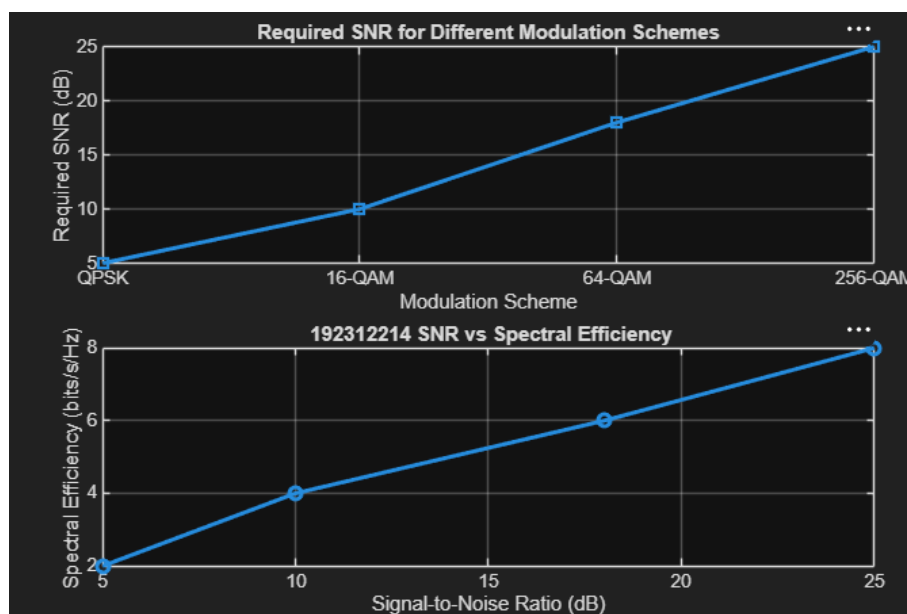
title('Required SNR for Different Modulation Schemes')

xlabel('Modulation Scheme')

ylabel('Required SNR (dB)')

grid on
```

Output



EXP 15: AI-based Traffic Prediction for 5G Networks

Objective 1: To simulate 5G network traffic data and visualize the traffic pattern.

Program code:

```
clc; clear; close all;

time = 1:20;

traffic = [20 25 30 35 40 50 60 58 65 70 ...
          75 80 78 85 90 95 100 98 105 110];

figure;

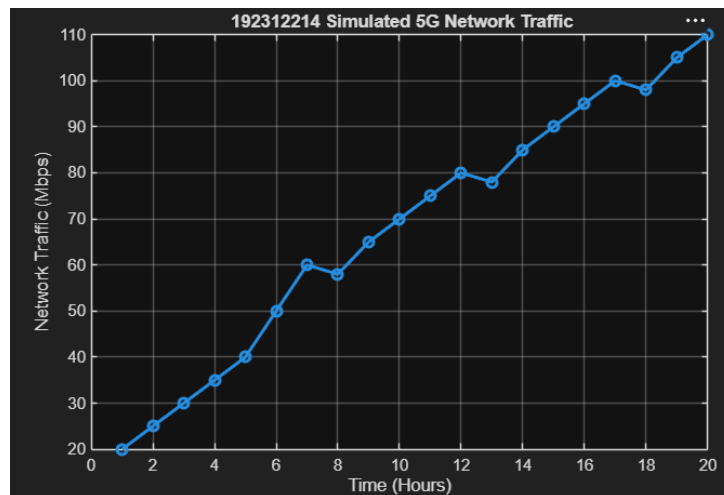
plot(time, traffic, '-o', 'LineWidth', 2); grid on;

xlabel('Time (Hours)');

ylabel('Network Traffic (Mbps)');

title('192312214 Simulated 5G Network Traffic');
```

Output:



Objective 2: To predict future 5G network traffic using MachineLearning (Linear Regression).

Program code:

```
clc;

clear;

close all;

% Simulated Traffic Data

time = 1:20;

traffic = [20 25 30 35 40 50 60 58 65 70 ...
          75 80 78 85 90 95 100 98 105 110];

% Linear Regression Model

model = polyfit(time, traffic, 1);
```

```

% Predicted Traffic

predicted = polyval(model, time);

figure;

plot(time, traffic, 'bo-', 'LineWidth', 2);

hold on;

plot(time, predicted, 'r*-', 'LineWidth', 2);

grid on;

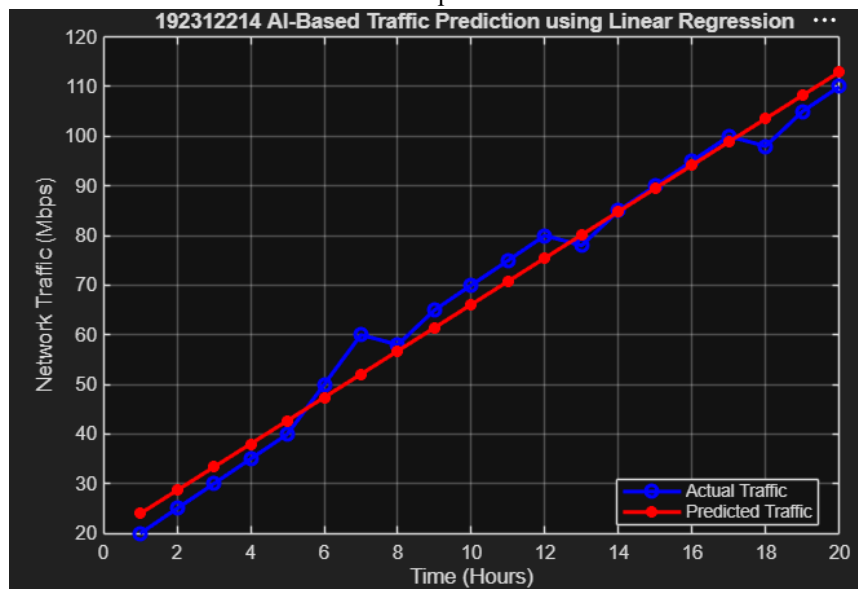
xlabel('Time (Hours)');

ylabel('Network Traffic (Mbps)');

title('192312214 AI-Based Traffic Prediction using Linear Regression');
legend('Actual Traffic', 'Predicted Traffic', 'Location', 'best');

```

Output:



Objective 3: To evaluate the prediction accuracy by comparing actual and predicted traffic values.

Program code:

```

clc;

clear;

close all;

% Simulated Traffic Data

time = 1:20;

traffic = [20 25 30 35 40 50 60 58 65 70 ...

    75 80 78 85 90 95 100 98 105 110];

model = polyfit(time, traffic, 1);

```

```

error = traffic - predicted;

RMSE = sqrt(mean(error.^2));

disp(['Root Mean Square Error (RMSE) = ', num2str(RMSE)]);

subplot(2,1,1);

plot(time, traffic, 'bo-', 'LineWidth', 2);

plot(time, predicted, 'r*- ', 'LineWidth', 2); grid on;

xlabel('Time (Hours)');

ylabel('Traffic (Mbps)');

title(' 192312214 Actual vs Predicted Traffic');

legend('Actual', 'Predicted', 'Location', 'best');

subplot(2,1,2);

bar(time, error); grid on;

xlabel('Time (Hours)');

ylabel('Prediction Error (Mbps)');

title(' 192312214 Prediction Error');

disp('Time Actual Predicted');

disp([time' traffic' predicted']);

Output:

```

